

# R1 Mock 2 — OOP keywords + cache

20 minutes · Ojas Srivastava · Intern – Technology

No notes. Faster interviewer. Twenty minutes is tight.

**From reports:** GFG SDE-1 R2: access specifiers, static, bank design. TalentBattle: favourite DS. GFG intern / SDE: array + map. Resume-driven Redis follow-up.

*Interviewer tone: jumps topics. Answer completely on the first pass.*

**How to use.** Set a 20-minute timer. Cover the green boxes. Answer out loud.

**Do NOT volunteer:** location, IFFCO (unless they open resume), MCP names, AirHelp, “I led the whole team.”

**Locked intro — say verbatim, then stop:**

Good afternoon sir, ma’am. I am Ojas Srivastava. Pre-final B.Tech AI, SVNIT Surat, CGPA 9.20. Coursework: OOP, DBMS, OS, CN, DAA. Google Solution Challenge 2026 Global rank 106 — LogiFlow, supply-chain routing. Vibe2Ship 2026 Global Top 20 — Community Hero, civic issue reporting PWA. Coding workflow: Cursor, MCP and plugins; spec and review every merge. Chairperson Nexus SVNIT, CSE+AI cell, manage and mentor 500+ students. I own the outcome. GEP works on AI-powered procurement and supply chain — operational data to a decision, owned till it ships.

## 1. Clock

|                    |          |                                                      |
|--------------------|----------|------------------------------------------------------|
| <b>0:00–2:00</b>   | Intro    | Locked intro, 60 seconds. Then silence.              |
| <b>2:00–8:00</b>   | OOP      | Access specifiers, static, abstraction, bank design. |
| <b>8:00–14:00</b>  | LogiFlow | Redis internals, algorithms, favourite DS = graph.   |
| <b>14:00–18:00</b> | DSA      | Two-sum with a map. One pass.                        |
| <b>18:00–20:00</b> | Close    | Why GEP.                                             |

## 2. Script

**0:00 — Intro**

**Interviewer.** *Hi Ojas. Quick intro — 60 seconds.*

**You say.** Locked intro, same content, slightly tighter pace. Still hit LogiFlow, Community Hero, Nexus, GEP bridge. Stop.

**2:00 — OOP keywords**

**Interviewer.** *Access specifiers — public, private, protected. Where do you use private?*

**You say.** Fields private; behaviour through methods. LogiFlow: cache keys and DB handles not exposed — callers use `getRoutes()` only.

**Interviewer.** *What does static mean? Same as constant?*

**You say.** Belongs to class, not instance. Static method cannot touch instance fields. **Not** constant.

**Interviewer.** *Abstraction vs encapsulation — people confuse these.*

**You say.** Abstraction hides complexity (what vs how). Encapsulation bundles data and restricts access. Different ideas.

**Interviewer.** *Design a bank account in OOP — 90 seconds, classes only.*

**You say.** Abstract Account: balance, `deposit()`, `withdraw()`. Savings/Current override withdraw rules. Transaction for audit. Balance private.

**8:00 — LogiFlow (resume-driven)**

**Interviewer.** *Redis on your resume. Explain exactly how you used it.*

**You say.** Cache-aside on hot corridor reads. Key from query params. TTL. Invalidate on write. SQL source of truth.

**Interviewer.** *Two users, same corridor, both cache miss — problem?*

**You say.** Both recompute — wasteful but OK for reads. Thundering herd possible; could add lock or single-flight. Not inventory-critical like cart stock.

**Interviewer.** *Which algorithms? Do not say ML unless you can defend it.*

**You say.** Graph over corridor network. BFS when hops uniform; weighted shortest path when not. Scoring on time, cost, delay.

**Interviewer.** *Favourite data structure?*

**You say.** Graph — stations nodes, corridors edges. Adjacency list (sparse). BFS for unweighted / level-order. Stay on graph if they grill internals.

**14:00 — DSA**

**Interviewer.** *Array and target — return indices of two numbers that sum to target. One pass.*

**You say.** For each  $x$ , check  $\text{target} - x$  in map **before** inserting  $x$ . Avoids using same element twice.  $O(n)$  time,  $O(n)$  space.

**Why check before insert?** For  $[3,3]$ , target 6 — cannot use index 0 twice.

**18:00 — Close**

**Interviewer.** *Why GEP?*

**You say.** Procurement + supply chain — operational data to a decision. Same shape as LogiFlow. SMART, NEXXE, Quantum Intelligence — product plus consulting judgement.

**3. Score this mock**

Rate each area /10. Pass bar: **7+** average.

|                 |                                                                |       |
|-----------------|----------------------------------------------------------------|-------|
| <b>Intro</b>    | Under 60s? Stopped? No location / IFFCO / MCP names?           | _____ |
| <b>OOP</b>      | Static $\neq$ constant? Bank design under 90s?                 | _____ |
| <b>LogiFlow</b> | SQL is source of truth — not “Redis is fast”? Graph, not tree? | _____ |
| <b>DSA</b>      | Check map before insert? Complexity stated?                    | _____ |
| <b>Close</b>    | Why GEP in two sentences, no stipend talk?                     | _____ |

**Pass bar.** Bank design under 90 seconds. Redis answer shows SQL-as-truth.

Next: GEP\_R1\_Mock\_3.pdf — morning of interview, no notes.